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## Effects of Strength Training on Important Determinants of Endurance Performance

Hickson et al. [1] were amongst the first to give us the idea that heavy strength training could positively affect cycling performance. Their study was performed on untrained individuals, but lately the knowledge of the effects of adding strength training to the endurance training in recreationally, well-trained and elite cyclists has increased. This chapter will mainly focus on the effect of adding strength training to the ongoing endurance training in well-trained cyclists. To do this, the effects of strength training on important physiological determinants of performance will be evaluated. The interaction between maximal oxygen consumption ( $\text{VO}_{2\text{max}}$ ), fractional utilization of  $\text{VO}_{2\text{max}}$  ( $\%\text{VO}_{2\text{max}}$ ), and the anaerobic capacity explains how long a given rate of aerobic and anaerobic metabolism can be sustained, which altogether determines the performance  $\text{VO}_2$  [2]. The cycling economy or efficiency then determines the power output at a given amount of energy consumption and thus majorly determines the performance power output [2]. We will now look closer on how strength training affects these performance determinants and performance power output.

Combining heavy strength training with normal endurance training seems to have neither a positive nor a negative effect on the development of  $\text{VO}_{2\text{max}}$  [3–15]. When it comes to gross efficiency or cycling economy, the findings are more equivocal. When cycling economy is measured by the traditional method (i.e., short, 3–5 min, submaximal bouts of cycling), no additive effect of supplemented strength training was observed in well-trained and elite cyclists [3, 10–12]. However, by using the same approach to measure cycling economy, improvements were shown

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after adding heavy strength training in moderately trained cyclists [4, 9, 14]. Interestingly, during 3 h submaximal cycling in well-trained male and female cyclists no benefit of heavy strength training on cycling economy was observed during the first hour, but during the last hour there was an improvement in cycling economy in the strength training group [16, 17]. The latter indicates that also well-trained cyclists may improve cycling economy when measured during prolonged, and maybe more ecologically valid, cycling.

In order to calculate the fractional utilization of  $\text{VO}_{2\text{max}}$ , a common practice is to use the percentage of  $\text{VO}_{2\text{max}}$  at the lactate threshold. Unlike the impression of the name, there is in practice no single unique lactate threshold, as it can be calculated in numerous ways [18]. However, in order to measure training-induced changes, it is important to use the same calculation method before and after the training period [19]. In theory, the lactate threshold describes an estimation of a breakpoint on the blood lactate concentration ( $[\text{La}^-]$ ) curve as a function of exercise intensity [19]. Lactate threshold expressed as a percentage of  $\text{VO}_{2\text{max}}$  is largely unaffected by exercise economy and  $\text{VO}_{2\text{max}}$ , which might explain the small correlation between lactate threshold expressed as  $\%\text{VO}_{2\text{max}}$  and time trial cycling performance in cyclists [20]. The few studies that have reported the effect of concurrent training on lactate threshold expressed as  $\%\text{VO}_{2\text{max}}$  in cyclists observe neither a positive nor negative effect [12, 14]. However, we recently reported that concurrent training in well-trained female cyclists improved fractional utilization of  $\text{VO}_{2\text{max}}$  during a 40-min all-out trial, while no changes occurred in the group that performed endurance training only [15]. Interestingly, in the same study, no significant change was observed when the lactate threshold was expressed as  $\%\text{VO}_{2\text{max}}$ , indicating that this method of expression fractional utilization of  $\text{VO}_{2\text{max}}$  might not be sensitive enough to detect small changes [15].

There are numerous methods to determine the power output at the lactate threshold, resulting in diverse “thresholds” on the  $[\text{La}^-]$  vs. power curve which all seem to correlate well with long-term endurance performance [18, 19]. The power output at the lactate threshold is, amongst others, affected by the cycling economy. Therefore, the finding of improved lactate threshold power output in several studies after combined heavy strength- and endurance training is expected [10–13, 15, 21]. However, there are also studies observing no improvements in power output at a defined  $[\text{La}^-]$  [3, 14, 22].

The traditional way of measuring cycling performance is to perform all-out tests lasting between 30 and 60 min, where mean power output typically is the performance measure. When positive effects of concurrent training are reported, heavy strength training is performed with multiple leg exercises during a period of minimum 8 weeks [3, 7, 11–13, 15, 17, 21]. In contrast, studies failing to show much improvement in measurement of performance were typically either short term in duration, included a low volume of strength training or used explosive strength training [5, 6, 8, 22]. A measurement which can be looked at as a performance measurement is the power output at  $\text{VO}_{2\text{max}}$  ( $W_{\text{max}}$ ), which is influenced by  $\text{VO}_{2\text{max}}$ , cycling economy, anaerobic capacity, and neuromuscular characteristics [23]. Accordingly,  $W_{\text{max}}$  has been shown to predict endurance performance in cyclists [24,

[25] and to distinguish the endurance performance in well-trained cyclists [25]. Concurrent endurance and heavy strength training are reported to increase  $W_{\max}$  or time to exhaustion at  $W_{\max}$  in trained to well-trained cyclists [7, 10–12, 14, 26]. A relatively new method of assessing cycling performance in the lab is an attempt to imitate road cycling competitions, by performing prolonged submaximal cycling for many hours followed by a short, 5-min, all-out trial which is the main performance outcome. Combining heavy strength training with usual endurance training has improved 5-min all-out power after 3 h submaximal cycling in both well-trained female and male cyclists [16, 17].

Another factor important for the outcome of a cycling race is the ability to close a gap, break away from the pack, or perform well in the final sprint. The outcome of these crucial moments of a race is largely decided by the size of the involved muscle mass and the maximal leg strength [27]. The ability to generate high power output for a short period is often measured in the lab as mean and peak power output in a 30-s all-out test. Based on the beneficial effects of heavy strength training on muscle strength and muscle mass, it is expected that concurrent training improves the ability to generate a high power output for a short period of time [10–13]. A related variable is the curvature constant ( $W'$ ) of the power–duration relationship and it has recently been observed that  $W'$  has a strong positive relationship with both thigh muscle size and maximal knee extensors force in elite cyclists [28]. This has practical implications, since the ability to generate high power output during a short period of time is an important aspect of overall cycling performance [29].

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## Potential Mechanisms Behind the Effects of Strength Training on Cycling Performance

There are multiple potential mechanisms behind the positive effects on cycling performance of adding heavy strength training to the ongoing endurance training. A potential mechanism for improved performance after combined strength and endurance training is an increased force potential in the efficient muscle fiber type I and thus postponed activation of the less efficient type II muscle fibers, resulting in improved cycling economy and performance [30]. Since mainly type I fibers are activated during traditional submaximal measurements of cycling economy, this might explain why the literature is equivocal on improvements in cycling economy in well-trained cyclists. Postponed activation of type II muscle fibers is a plausible explanation for the findings of improved cycling economy in well-trained cyclists after a period of concurrent training after 2 h of submaximal cycling [16, 17]. It is likely that prolonged cycling exhausts some of the type I fibers and therefore can postpone activation of the less efficient type II fibers explain improved cycling economy. The latter is in agreement with the observation of strength training induced reduction in the increased muscle activity during the second hour of a 2-h cycling test [31], indicating postponed activation of type II fibers. The latter may have a glycogen sparing effect that might contribute to explain improved 5-min all-out performance after 3 h submaximal cycling after

12 weeks of concurrent training [16, 17]. Another potential mechanism related to muscle fiber recruitment is an increased proportion of type IIA fibers at the expense of type IIX fibers. It has been observed increased proportion of type IIA fibers at the expense of type IIX fibers after 12–16 week heavy strength training in both well-trained female and top-level male cyclists [3, 15]. The increase in the more fatigue resistant, yet high capability of power output, type IIA fibers may contribute to improved endurance performance. Indeed, recent data from our lab reveals a large correlation between change in mean power output during a 40-min all-out test and change in IIX fibers ( $r = -0.63$ ,  $p < 0.05$ ). This means that reduced proportion of type IIX fibers after 12 weeks of heavy strength training (and endurance training) was associated with improved power output during a 40-min all-out trial in well-trained female cyclists [15].

Another putative mechanism explaining improvement in endurance-related measurements after concurrent training is increased maximum force, and/or increased rate of force development (RFD) facilitating better blood flow to exercising muscles [3, 14, 32]. Improvement in maximum force and/or RFD might lower the relative exercise intensity and induce less constriction of the blood flow. Alternatively, improved RFD may reduce time to reach the desired force in each movement cycle and thereby potentially increase the relaxation phase with improved blood flow. Interestingly, 25 weeks of combined strength and endurance training in elite cyclists led to earlier occurrence of peak torque during the pedal stroke, while endurance training only did not [12]. Whether blood flow is enhanced after a period of concurrent training has not been thoroughly investigated, but in theory will increased blood flow lead to increased delivery of  $O_2$  and substrates to working muscles and thus contribute to increased power output at a fixed  $[la^-]$ , but not necessarily improved cycling economy. Accordingly, it has been observed that improvement in power output at  $4 \text{ mmol} \cdot \text{L}^{-1} [la^-]$  and mean power in 40-min all-out trial correlated largely with changes towards earlier peak torque during the pedal stroke ( $r = -0.50$  and  $r = -0.63$ , respectively,  $p < 0.05$ ; [12]). On the other hand, a recent study on moderately trained cyclists by Barrett-O'Keefe et al. [4] showed that 8 weeks of heavy strength training improved work economy at a cadence of 60 rpm, reduced muscular blood flow, while maintaining muscular arterial-venous oxygen difference. The latter indicates that improvement in muscular efficiency is an important mechanism behind improved work economy and improved endurance performance.

Coyle et al. [33] observed a positive relationship between lean body mass and 1-h all-out power output in well-trained and elite cyclists, suggesting that improved ability to recruit a relatively large quantity of muscle mass in each pedal stroke was associated with elite performance. Furthermore, cyclists that use a larger amount of their muscles mass have been observed to have a larger fractional utilization of  $VO_{2\max}$  [33]. This was related to increased amounts of mitochondria sharing the power production and leading to less metabolic strain at a certain power output. Increased fractional utilization was actually indicated in a study where elite cyclists improve 45-min all-out performance after 16 weeks of added heavy strength training. The authors estimated that the power output during the

test had increased from 76% to 83% of the power output at  $\text{VO}_{2\text{max}}$  [3]. It might thus be speculated that increased lean mass and increased absolute amounts of aerobic enzymes, with no change in the concentration of aerobic enzymes, could give a contribution to improved performance after adding heavy strength training to normal endurance training. This is in agreement with the recent results from our lab, where there was a very large correlation between change in mean power output during the 40-min all-out test and change in the cross-sectional area of m. quadriceps ( $r = 0.73$ ,  $p < 0.05$ ) with no change in the concentration of aerobic enzymes [15]. Furthermore, in the later study, fractional utilization of  $\text{VO}_{2\text{max}}$  during the 40-min all-out test improved larger in the concurrent group than the control group [15].

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## Practical Recommendations for Implementing Strength Training

In this chapter, we have shown that adding heavy strength training to the normal endurance training can have a positive effect on cycling performance, at least when it is performed above 8 weeks, with multiple exercises for the power generating muscles in cycling. Nevertheless, it is important to remember that endurance training on the bike is the most important focus for a cyclist. If you normally have only three training sessions available during the week, is it likely that you will have the largest performance increments they are spent on the bike. However, if more time is available, then the scientific literature indicates that adding some heavy strength training *can* improve your cycling performance.

When choosing strength training exercises, you should have the specificity principle in mind. That means that the muscle action, the muscles engaged, and the movement pattern should be somewhat similar to the action on the bike. Specificity is recommended partly due to adaptations in the neural system (like optimal activation of the involved muscles) and partly due to structural adaptations (like optimizing the number of active cross-bridges in the particular range of motion). The latter is indicated by findings from an isometric strength training study where the strength improvement at the trained muscle length was twice the improvement of other muscle lengths [34]. The major contribution to the power output during cycling is achieved from concentric muscle actions during the pedaling down stroke. Peak force during pedaling occurs when the crank arm reaches an angle of approximately  $90^\circ$ , which is usually equal to a knee angle of approximately  $100^\circ$ . Therefore, a general rule is to focus on strength training exercises with a knee angle between  $90^\circ$  and almost full knee extension. That being said, the exercises should not be so difficult to perform that the mechanical loading and muscle mass involved is too low. Mechanical loading is an important stimulus for strength training adaptations. For instance, it has been shown that what somebody call “strength training” on the bike or “power pedaling” which consists of cycling with a low cadence (around 40 rpm) with relatively high force has no effect on neither maximal force capacity of the legs nor cycling performance [35].

It seems like it is the intended rather than the actual velocity that determines the velocity-specific training response [36]. This means that even though the actual movement velocity is quite low, you may increase your RFD if you focus on performing the lift as quick as possible. Therefore, you are recommended to have maximal effort in the concentric, cycling specific phase; performing the concentric phase as quickly as possible, while the eccentric, non-cycling specific phase should be performed more slowly (lasting around 2–3 s).

In terms of training periodization, it is recommended to build up the maximal strength during the first phase of the preparatory period leading up to the competition season. Two strength training sessions per week are normally enough to achieve a sufficient increase in strength during a 8–12 week period [10, 13–15]. We have seen an increase in maximum strength of 23–26% after 10–12 weeks with a strength training program designed as a “daily undulating periodized program” with progression in intensity, starting with 10 repetition maximum (RM) and ending at 4RM. This program varies the training load from session to session and progresses towards heavier loads and fewer repetitions. It is recommended to perform between 4RM and 10RM and 2–3 sets with approximately 2–3 min of rest between sets. Before you start with heavy loads, you must ensure that you have first developed a proper lifting technique with lower loads.

Note that in the beginning of the strength training period, it is common to get “heavy” and “sore” legs after the strength training sessions. Therefore, it is important to take it easy with the endurance training during the first two to 3 weeks of the strength training. It may be a good advice to start strength training rather quickly after the end of a competition season, when endurance training has a lower priority. Some potential exercises to choose amongst when designing your strength training program can be: half squat, single leg half squat, step-up, leg press with one foot at a time, one-legged hip flexion (imitating the pedaling upstroke), and toe raises (to ensure proper force transmission from the large thigh muscles into the pedal). You should perform 5–10 min of general warm-up followed by a specific warm-up with a gradual increase in loading of the strength training exercises. It is recommended to start the session with the exercise that involves the largest muscle mass, often the most coordinative demanding exercise. Thereafter, complete 2–3 more exercises that focus on the important muscles for the pedaling action. Strength training sessions to increase cycling performance do not have to be time consuming, including warm-up which can be done within 45 min.

There are phases during the preparatory period where you want to increase the focus on endurance training. In these phases, you should try to maintain the strength training adaptations. It seems like this can be done by performing one heavy strength training session every 7–10th day [11, 12]. To avoid detraining effects, it is recommended that you perform high intensity muscle actions and maximal mobilization in the concentric phase, but you do not need to follow the repetition maximum principle. That means that you for example can perform  $2\text{--}3 \times 5$  repetitions at a load which you can perform maximally 8–10 repetitions. It is also recommended to perform strength maintenance training during the competition season. We have recently seen that if you stop all strength training from the middle of April until the middle of June, you will lose almost all your strength training adaptations [37].

## Summary

This chapter provides an updated review on the effects of adding heavy strength training to the ongoing endurance training on the major physiological determinants of endurance performance in well-trained cyclists. Although not universal, the scientific literature present findings of improved cycling economy, improved fractional utilization of  $\text{VO}_{2\text{max}}$ , and improvement in indices of anaerobic power after adding heavy strength training to well-trained cyclists. More important, and more consistent, is the finding of improved cycling performance and/or improved  $W_{\text{max}}$  or time to exhaustion at  $W_{\text{max}}$  after adding heavy strength training with multiple exercises targeting the important muscles for power output during the pedal stroke for a longer duration (above 8 weeks).

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